



**Topic Test: OxfordAQA
International GCSE Chemistry 9202**

Trends within the periodic table

Name: _____

Class: _____

Date: _____

Time: **50 minutes**

Marks: **50 marks**

Comments:

1

Platinum and gold can both be used to make wedding rings.



By Jeff Belmonte from Cuiabá, Brazil (Flickr) [CC-BY-2.0], via Wikimedia Commons

Use the Data Sheet to help you to answer these questions.

- (a) Draw a ring around the part of the periodic table in the list below to which platinum and gold belong.

group 1

group 2

transition elements

group 7

(1)

- (b) Platinum and gold have properties that make them suitable for wedding rings.

Tick (✓) **two** of these properties.

Property	Tick (✓)
These metals do not react with air.	
These metals have low melting points.	
These metals do not react with water.	
These metals have low densities.	

(2)

(Total 3 marks)

(a) The periodic table on the Data Sheet may help you to answer these questions.

[illegible]

Which letter, **A**, **B**, **C**, **D** or **E**, represents:

11

(1)

11

(1)

11

(1)

11

(1)

- (b) A chemistry teacher demonstrated the reaction between sodium and water to some students. One of the students wrote the following notes.

The reaction between sodium and water

A piece of sodium was cut easily into smaller pieces with a knife.

The sodium was added to water in a trough.

The sodium:

- floated
- melted quickly to give a silvery ball
- moved on the surface of the water
- fizzed.

Use the information in the box to help you to answer these questions.

What evidence is there that:

- (i) sodium has a low melting point

(1)

- (ii) sodium is soft

(1)

- (iii) a gas was produced?

(1)

(Total 7 marks)

3

(a) Part of the periodic table is shown below.

[illegible]

Choose your answers **only** from the letters shown in the periodic table above.

(i) aluminium

7

(1)

(ii) a Group 5 element

7

(1)

(iii) an alkali metal

7

(1)

(iv) the element with atomic (proton) number of 47

10

(1)

(v) an element with seven electrons in its outer shell?

(1)

(b) The table shows the boiling points of the Group 7 elements.

The elements are arranged in **alphabetical order**.

Group 7 element		
Name	Symbol	Boiling point in °C
Astatine	At	337
Bromine		58
Chlorine	Cl	-34
Fluorine	F	-188
Iodine	I	184

(i) The symbol for bromine is missing from the table.

What is the symbol for bromine?

Symbol = _____

(1)

(ii) Arrange these elements in order of **decreasing** boiling point. The first one and the last one have been done for you.

At _____ _____ _____ F

Highest boiling point  Lowest boiling point

(1)

- (c) The table shows some statements about Group 7 elements.

Tick (✓) the **two** correct statements.

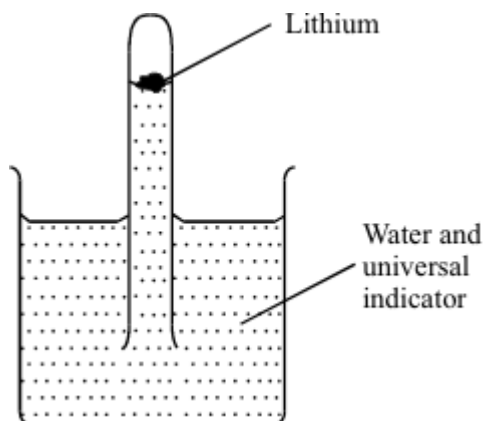
	Tick (✓)
They are halogens.	
They are metals.	
They become less reactive down Group 7.	
They are compounds.	

(2)

(Total 9 marks)

4

The diagram shows an experiment to study the reaction of lithium with water.



- (a) Describe, as fully as you can, what you would see as the lithium reacts with the water in this experiment.

To gain full marks in this question you should write your ideas in good English. Put them into a sensible order and use the correct scientific words.

(3)

- (b) The reaction has two products. Complete the word equation for this reaction by choosing the correct substances from the box.

hydrogen	lithium hydride	lithium hydroxide
lithium oxide		oxygen

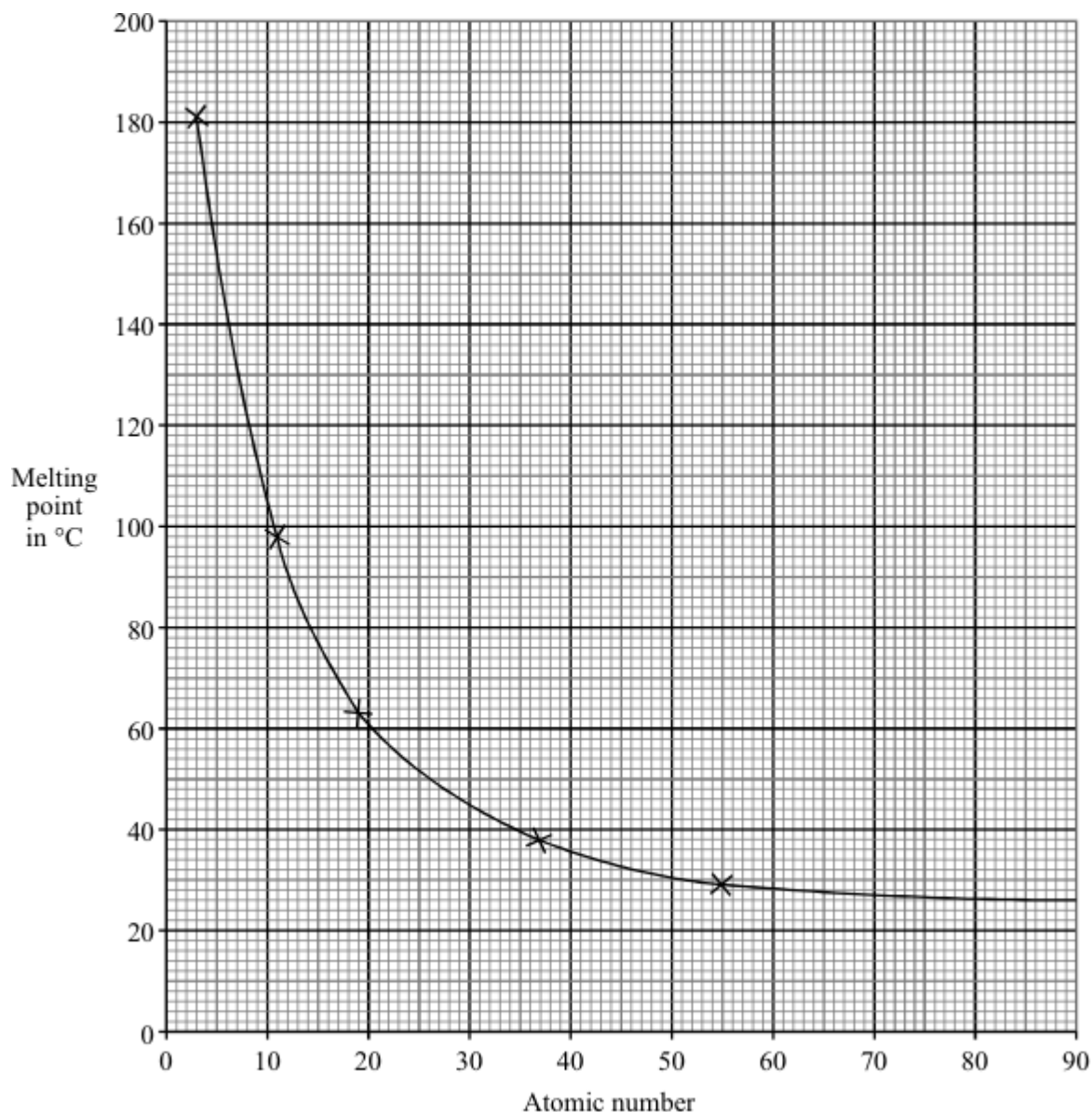
lithium + water → _____ + _____

(2)

- (c) Caesium is lower down in Group 1 of the periodic table than lithium.
Suggest how the reaction of caesium with water might be different from lithium's reaction.

(1)

- (d) The graph shows the melting points of the Group 1 metals plotted against their atomic numbers.



- (i) Describe fully how the melting points change as the atomic number increases.

(2)

- (ii) Francium has an atomic number of 87.
Use the graph to estimate the melting point of francium.

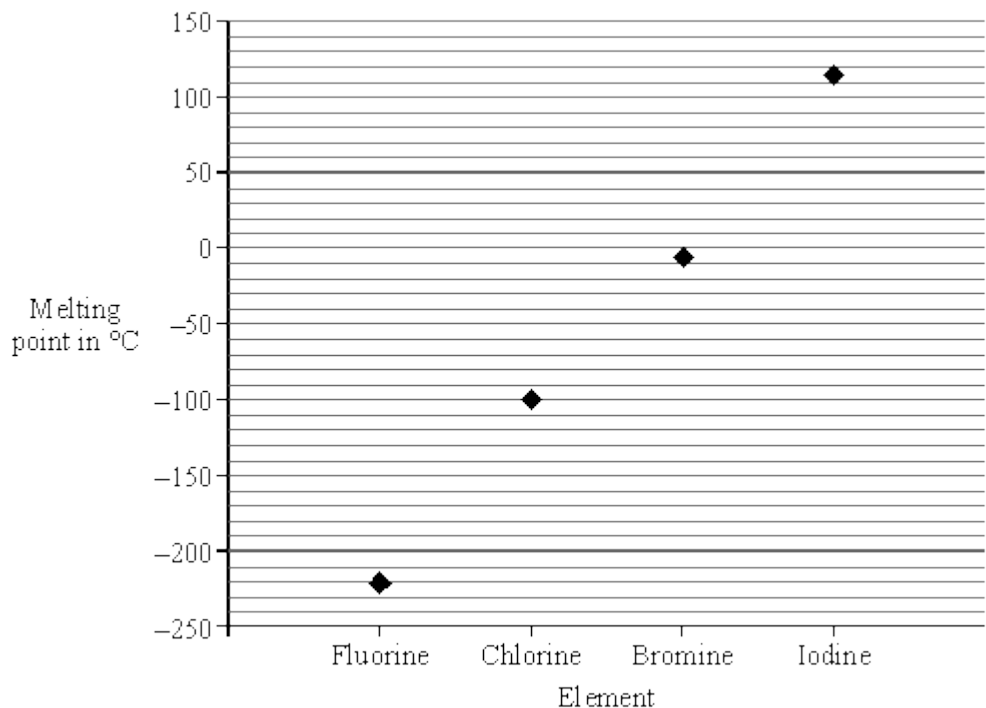
Estimate of melting point _____ °C

(1)

(Total 9 marks)

5

The graph shows the melting point of four elements in Group 7 of the periodic table.



(a) What is the melting point of fluorine?

(1)

(b) Room temperature is 20°C.

Which element is solid at room temperature?

(1)

(c) Look at the periodic table on the Data Sheet.

Using data from the graph, describe the trend of melting points of the elements in Group 7.

(2)

- (d) The elements in Group 7 are non-metals.

Which **two** of the following are properties of non-metals?

Place a tick (✓) in the box against each correct property.

Brittle (if solid)

☐

Good conductor of heat

☐

High boiling point

☐

Poor conductor of electricity

☐

(2)

(Total 6 marks)

6

Sodium and potassium are both in Group 1 of the Periodic Table.

- (a) Explain, by reference to their electronic structures, why both elements are placed in Group 1.

(1)

- (b) Use the Data Sheet to help you to answer this question.
The diagrams below represent the electronic structures of some atoms and ions.



A



B



C



D



E



F

Which one of the structures, **A - F**

- (i) represents a sodium **atom**, _____

(1)

- (ii) represents a potassium **ion**? _____

(1)

- (c) Sodium and potassium both react with cold water.

- (i) The word equation represents the reaction of sodium with water.

sodium + water → sodium hydroxide + hydrogen

Complete and balance the symbol equation for this reaction.



(2)

- (ii) How does the reactivity of potassium with water differ from that of sodium with water?

Explain this difference in reactivity by reference to the electronic structures of the potassium and sodium atoms.

(4)

(Total 9 marks)

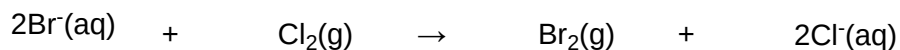
7

The halogens are in Group 7 of the periodic table.

- (a) Why, in terms of electrons, are the halogens in Group 7?

(1)

- (b) Sea water contains bromide ions (Br^-).
The bromide ions can be changed to bromine by bubbling chlorine gas into sea water.
Chlorine is able to displace bromine from sea water because chlorine is more reactive than bromine.



Explain, in terms of electrons, why chlorine is more reactive than bromine.

(3)

(Total 4 marks)

8

- (a) What is the name given to the block of elements in the middle of the Periodic Table which includes vanadium?

(1)

(b) Some of the properties of vanadium are shown in this list.

- It has a high melting point.
- It is a solid at room temperature.
- It is a conductor of electricity.
- It is a good conductor of heat.
- It forms coloured compounds.
- It forms crystalline compounds.
- It forms compounds that are catalysts.

Select **two** properties, from the list above, which are **not** typical of a Group 1 metal.

1. _____

2. _____

(2)

(Total 3 marks)

Mark schemes

1	(a) transition elements	1	[3]
	(b) These metals do not react with air	1	
	These metals do not react with water	1	
2	(a) (i) E	1	[7]
	(ii) B	1	
	(iii) C	1	
	(iv) A	1	
	(b) (i) quickly melted <i>allow melts in contact with water,</i> <i>allow bp 100 °C (of water) shows mp is low</i> <i>ignore one other piece of information</i>	1	
	(ii) easily cut <i>ignore one other piece of information</i>	1	
	(iii) effervescence / fizzing / bubbling <i>ignore named gas</i> <i>ignore one other piece of information</i>	1	
3	(a) (i) B	1	
	(ii) E	1	
	(iii) F	1	
	(iv) D	1	
	(v) C	1	

(b) (i) Br

do not accept BR or br or bR
ignore numbers
allow written in table if answer blank

1

(ii) I Br Cl

allow iodine, bromine, chlorine
allow I,B,C
allow capitals or lower case
allow 184, 58, -34

ignore numbers

1

(c) they are halogens

1

they become less reactive down Group 7

1

[9]

4

(a) **Quality of Written Communication**

The answer to this question requires ideas in good English in a sensible order with correct use of scientific terms. Quality of written communication should be considered in crediting points in the mark scheme.

maximum 2 marks if ideas not well expressed

any **three** from:

*do **not** accept flames*

floats

fizzes / bubbles or produces a gas

ignore reference to heat

indicator goes blue / purple / violet (alkaline colour)

3

water level in test tube goes down **or** gas fills the test tube

lithium 'dissolves' (owtte)

moves around (on surface of water)

steam

(b) lithium hydroxide

1

hydrogen

1

- (c) more violent / reactive
accept a description of the reaction which indicates greater violence 1
- (d) (i) decreases 1
- and then slows down **or** levels off 1
- (ii) 26(°C) 1

[9]

5

- (a) -220(°C)
accept without unit
*do **not** credit if no minus sign*
no tolerance allowed 1
- (b) iodine
allow I or I₂ 1
- (c) increase down (the groups)
*N.B. must be **one** comparative or superlative for either position in group or melting point to obtain 2 marks*
e.g. accept lowest in group has highest melting point for 2 marks
accept highest in group has lowest melting point for 2 marks
accept highest mass numbers or proton numbers or atomic numbers have highest melting point for 2 marks
*allow F has lower **or** lowest melting point for 1 mark*
allow 'Increase from F to 1', i.e. referring to graph rather than periodic table for 1 mark
F has a low melting point no marks 2
- (d) brittle 1
- poor conductor of electricity
maximum 1 mark if three boxes ticked
no marks if all four boxes ticked 1

[6]

6

- (a) same number of electrons in outer shell / 1 / an electron in outer shell / lose one electron
for 1 mark 1
- (b) (i) C
for 1 mark 1
- (ii) E
for 1 mark 1
- (c) (i) $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$
symbols must be correct
correct multiples / fractions accepted
Balancing mark is independent
formulae gains 1 mark
balancing gains 1 mark 2
- (ii) Assume 'it' means potassium
potassium more reactive / vigorous / faster reaction / violent (can be awarded in either section)
potassium atom larger than sodium / higher outer energy level /
outer shell further from nucleus / more shells (not just more electrons)
electron in outer shell is less strongly attracted / greater shielding
outer electron more easily lost
for 1 mark each 4

[9]**7**

- (a) all have seven electrons in their outer shell / energy level 1
- (b) *must be comparative in all points or converse*
- chlorine atom is smaller than bromine atom
- or**
chlorine atom has fewer shells than bromine atom 1
- outer shell / energy level of chlorine has stronger (electrostatic) attraction to the nucleus than bromine
- or**
outer shell of chlorine is less shielded from the nucleus than bromine 1
- so chlorine more readily gains an extra electron 1

[4]

8

- (a) transition / transitional metals / elements / d-block
for one mark

1

- (b) coloured
catalyst

(accept high melting point)
for 1 mark each

2

[3]